Reversed achievement score in high school for the oldest child

Current 2022/2023 enrollment defines the high school sample. One oldest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.623	-0.072 [0.294] N=774	-0.013 [0.293] N=614	-0.047 [0.316] N=465	0.009 [0.343] N=324	0.362 [0.399] N=195
Social studies and life sciences	3.638	-0.048 [0.192] N=735	-0.003 [0.192] N=581	-0.092 [0.211] N=441	-0.032 [0.237] N=308	**0.420* [0.240] N=189**
Mathematics	3.359	-0.109 [0.340] N=761	-0.054 [0.348] N=603	-0.139 [0.372] N=456	-0.128 [0.415] N=317	0.060 [0.507] N=191
Science	3.529	0.089 [0.307] N=733	0.094 [0.313] N=580	0.041 [0.346] N=446	0.030 [0.368] N=312	0.195 [0.495] N=191
Foreign language	3.388	-0.496 [0.345] N=750	-0.566 [0.350] N=595	-0.601 [0.388] N=451	-0.368 [0.430] N=314	0.109 [0.479] N=191
Culture arts and sports	3.815	-0.133 [0.488] N=694	-0.064 [0.487] N=544	-0.085 [0.468] N=413	0.060 [0.565] N=284	0.257 [0.589] N=171
Vocational and technological	3.751	**-0.591** [0.264] N=550**	**-0.506* [0.291] N=434**	**-0.515* [0.308] N=323**	-0.494 [0.345] N=227	-0.133 [0.384] N=138

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.037 [0.335] N=758	0.054 [0.325] N=598	-0.001 [0.367] N=449	0.074 [0.403] N=308	0.542 [0.465] N=179
Social studies and life sciences	-0.200 [0.234] N=719	-0.126 [0.242] N=565	-0.272 [0.275] N=425	-0.217 [0.309] N=292	0.385 [0.311] N=173
Mathematics	-0.021 [0.437] N=746	0.061 [0.441] N=588	-0.032 [0.481] N=441	-0.015 [0.551] N=302	0.383 [0.719] N=176
Science	-0.093 [0.349] N=718	-0.069 [0.370] N=565	-0.162 [0.411] N=431	-0.153 [0.456] N=297	0.070 [0.646] N=176
Foreign language	-0.590 [0.411] N=736	-0.632 [0.413] N=581	-0.678 [0.459] N=437	-0.361 [0.507] N=300	0.209 [0.583] N=177
Culture arts and sports	-0.006 [0.601] N=680	0.125 [0.551] N=530	0.077 [0.532] N=399	0.315 [0.633] N=270	0.516 [0.672] N=157
Vocational and technological	**-0.871*** [0.292] N=540**	**-0.748** [0.299] N=424**	**-0.838** [0.321] N=313**	**-0.738* [0.376] N=217**	-0.259 [0.432] N=128

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.623	-0.043 [0.281] N=774	0.026 [0.277] N=614	0.006 [0.296] N=465	0.040 [0.325] N=324	0.328 [0.383] N=195
Social studies and life sciences	3.638	-0.023 [0.188] N=735	0.024 [0.185] N=581	-0.051 [0.202] N=441	0.000 [0.230] N=308	0.390 [0.237] N=189
Mathematics	3.359	-0.100 [0.334] N=761	-0.049 [0.344] N=603	-0.120 [0.365] N=456	-0.117 [0.409] N=317	0.041 [0.518] N=191
Science	3.529	0.112 [0.317] N=733	0.122 [0.319] N=580	0.078 [0.354] N=446	0.052 [0.375] N=312	0.165 [0.516] N=191
Foreign language	3.388	-0.478 [0.325] N=750	-0.536 [0.328] N=595	-0.574 [0.358] N=451	-0.376 [0.399] N=314	0.069 [0.462] N=191
Culture arts and sports	3.815	-0.125 [0.483] N=694	-0.059 [0.485] N=544	-0.086 [0.469] N=413	0.068 [0.553] N=284	0.241 [0.577] N=171
Vocational and technological	3.751	**-0.590** [0.265] N=550**	**-0.510* [0.289] N=434**	-0.515 [0.310] N=323	-0.481 [0.361] N=227	-0.124 [0.380] N=138

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.050 [0.327] N=758	0.058 [0.317] N=598	0.018 [0.352] N=449	0.076 [0.393] N=308	0.504 [0.469] N=179
Social studies and life sciences	-0.209 [0.217] N=719	-0.121 [0.230] N=565	-0.258 [0.254] N=425	-0.205 [0.294] N=292	0.344 [0.318] N=173
Mathematics	-0.024 [0.433] N=746	0.062 [0.440] N=588	-0.024 [0.475] N=441	-0.013 [0.547] N=302	0.342 [0.752] N=176
Science	-0.106 [0.341] N=718	-0.072 [0.362] N=565	-0.146 [0.405] N=431	-0.149 [0.453] N=297	0.025 [0.669] N=176
Foreign language	-0.602 [0.388] N=736	-0.626 [0.392] N=581	-0.671 [0.431] N=437	-0.377 [0.479] N=300	0.169 [0.565] N=177
Culture arts and sports	-0.006 [0.599] N=680	0.126 [0.551] N=530	0.077 [0.533] N=399	0.308 [0.629] N=270	0.491 [0.672] N=157
Vocational and technological	**-0.869*** [0.292] N=540**	**-0.740** [0.300] N=424**	**-0.836** [0.319] N=313**	**-0.741* [0.387] N=217**	-0.243 [0.439] N=128

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.623	-0.049 [0.289] N=774	0.022 [0.283] N=614	0.011 [0.302] N=465	0.063 [0.335] N=324	0.392 [0.393] N=195
Social studies and life sciences	3.638	-0.064 [0.193] N=735	-0.033 [0.193] N=581	-0.102 [0.218] N=441	-0.030 [0.251] N=308	**0.484* [0.245] N=189**
Mathematics	3.359	-0.091 [0.337] N=761	-0.043 [0.351] N=603	-0.143 [0.370] N=456	-0.007 [0.450] N=317	0.182 [0.613] N=191
Science	3.529	0.052 [0.316] N=733	0.055 [0.322] N=580	0.006 [0.352] N=446	0.031 [0.400] N=312	0.315 [0.534] N=191
Foreign language	3.388	**-0.587* [0.347] N=750**	**-0.621* [0.344] N=595**	-0.611 [0.371] N=451	-0.316 [0.401] N=314	0.093 [0.479] N=191
Culture arts and sports	3.815	-0.222 [0.489] N=694	-0.172 [0.488] N=544	-0.191 [0.473] N=413	-0.021 [0.547] N=284	-0.018 [0.570] N=171
Vocational and technological	3.751	**-0.530* [0.308] N=550**	-0.497 [0.336] N=434	-0.528 [0.370] N=323	-0.417 [0.440] N=227	-0.063 [0.418] N=138

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.048 [0.333] N=758	0.060 [0.322] N=598	0.041 [0.362] N=449	0.142 [0.409] N=308	0.714 [0.471] N=179
Social studies and life sciences	-0.241 [0.223] N=719	-0.166 [0.240] N=565	-0.288 [0.279] N=425	-0.201 [0.333] N=292	**0.582* [0.331] N=173**
Mathematics	-0.010 [0.438] N=746	0.073 [0.450] N=588	-0.049 [0.488] N=441	0.120 [0.613] N=302	0.515 [0.906] N=176
Science	-0.177 [0.336] N=718	-0.151 [0.357] N=565	-0.236 [0.396] N=431	-0.178 [0.483] N=297	0.245 [0.745] N=176
Foreign language	**-0.694* [0.415] N=736**	**-0.688* [0.410] N=581**	-0.667 [0.438] N=437	-0.245 [0.464] N=300	0.243 [0.556] N=177
Culture arts and sports	-0.104 [0.618] N=680	0.016 [0.567] N=530	-0.003 [0.543] N=399	0.272 [0.615] N=270	0.205 [0.690] N=157
Vocational and technological	**-0.727* [0.420] N=540**	-0.651 [0.421] N=424	-0.748 [0.472] N=313	-0.552 [0.566] N=217	-0.107 [0.553] N=128

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.623	-0.070 [0.272] N=774	0.026 [0.270] N=614	0.013 [0.290] N=465	0.083 [0.327] N=324	0.405 [0.378] N=195
Social studies and life sciences	3.638	-0.073 [0.190] N=735	-0.013 [0.187] N=581	-0.083 [0.209] N=441	-0.008 [0.239] N=308	**0.437* [0.255] N=189**
Mathematics	3.359	-0.159 [0.340] N=761	-0.108 [0.344] N=603	-0.233 [0.359] N=456	-0.121 [0.429] N=317	0.100 [0.570] N=191
Science	3.529	0.037 [0.309] N=733	0.069 [0.300] N=580	0.009 [0.330] N=446	0.039 [0.365] N=312	0.217 [0.520] N=191
Foreign language	3.388	**-0.627* [0.321] N=750**	**-0.671** [0.325] N=595**	**-0.713** [0.353] N=451**	-0.501 [0.382] N=314	-0.042 [0.476] N=191
Culture arts and sports	3.815	-0.217 [0.472] N=694	-0.112 [0.478] N=544	-0.154 [0.473] N=413	0.015 [0.555] N=284	0.064 [0.601] N=171
Vocational and technological	3.751	**-0.519* [0.307] N=550**	-0.446 [0.327] N=434	-0.444 [0.384] N=323	-0.312 [0.469] N=227	0.043 [0.525] N=138

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.086 [0.319] N=758	0.047 [0.309] N=598	0.016 [0.346] N=449	0.122 [0.383] N=308	0.551 [0.442] N=179
Social studies and life sciences	-0.266 [0.212] N=719	-0.167 [0.232] N=565	-0.299 [0.260] N=425	-0.215 [0.299] N=292	0.348 [0.337] N=173
Mathematics	-0.088 [0.438] N=746	0.006 [0.438] N=588	-0.132 [0.469] N=441	0.021 [0.568] N=302	0.454 [0.818] N=176
Science	-0.177 [0.330] N=718	-0.118 [0.339] N=565	-0.211 [0.374] N=431	-0.151 [0.441] N=297	0.056 [0.682] N=176
Foreign language	**-0.737* [0.385] N=736**	**-0.745* [0.388] N=581**	**-0.767* [0.413] N=437**	-0.436 [0.433] N=300	0.101 [0.567] N=177
Culture arts and sports	-0.107 [0.589] N=680	0.073 [0.538] N=530	0.010 [0.533] N=399	0.262 [0.627] N=270	0.285 [0.724] N=157
Vocational and technological	**-0.795** [0.362] N=540**	**-0.669* [0.355] N=424**	**-0.728* [0.418] N=313**	-0.542 [0.529] N=217	-0.025 [0.621] N=128

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.614	-0.014 [0.283] N=733	0.034 [0.275] N=583	-0.046 [0.303] N=444	0.016 [0.327] N=312	0.341 [0.400] N=187
Social studies and life sciences	3.631	-0.003 [0.190] N=694	0.026 [0.187] N=550	-0.114 [0.208] N=420	-0.027 [0.227] N=296	**0.416* [0.242] N=181**
Mathematics	3.351	-0.125 [0.336] N=720	-0.078 [0.345] N=572	-0.152 [0.368] N=435	-0.141 [0.415] N=305	-0.025 [0.550] N=183
Science	3.521	0.060 [0.323] N=693	0.069 [0.327] N=550	0.049 [0.351] N=425	0.067 [0.375] N=300	0.191 [0.549] N=183
Foreign language	3.381	-0.484 [0.330] N=710	-0.544 [0.331] N=564	-0.577 [0.362] N=430	-0.416 [0.380] N=302	0.070 [0.482] N=183
Culture arts and sports	3.814	-0.140 [0.485] N=659	-0.090 [0.490] N=517	-0.131 [0.476] N=394	0.008 [0.563] N=273	0.185 [0.610] N=163
Vocational and technological	3.750	**-0.636** [0.270] N=521**	**-0.588* [0.296] N=411**	**-0.562* [0.321] N=309**	-0.597 [0.391] N=218	-0.216 [0.414] N=131

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.004 [0.325] N=717	0.084 [0.307] N=567	-0.028 [0.353] N=428	0.071 [0.383] N=296	0.546 [0.484] N=171
Social studies and life sciences	-0.156 [0.224] N=678	-0.095 [0.232] N=534	-0.305 [0.265] N=404	-0.198 [0.293] N=280	0.408 [0.333] N=165
Mathematics	-0.043 [0.435] N=705	0.032 [0.441] N=557	-0.055 [0.480] N=420	-0.034 [0.558] N=290	0.264 [0.821] N=168
Science	-0.154 [0.352] N=678	-0.129 [0.374] N=535	-0.174 [0.409] N=410	-0.120 [0.457] N=285	0.064 [0.721] N=168
Foreign language	-0.580 [0.390] N=696	-0.612 [0.391] N=550	-0.655 [0.425] N=416	-0.395 [0.439] N=288	0.132 [0.570] N=169
Culture arts and sports	-0.005 [0.595] N=645	0.107 [0.549] N=503	0.046 [0.522] N=380	0.262 [0.633] N=259	0.437 [0.738] N=149
Vocational and technological	**-0.930*** [0.280] N=511**	**-0.837*** [0.283] N=401**	**-0.847** [0.338] N=299**	**-0.834* [0.445] N=208**	-0.267 [0.514] N=121

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.614	-0.020 [0.282] N=733	0.026 [0.275] N=583	-0.060 [0.300] N=444	0.008 [0.326] N=312	0.342 [0.389] N=187
Social studies and life sciences	3.631	-0.015 [0.194] N=694	0.016 [0.190] N=550	-0.124 [0.209] N=420	-0.040 [0.228] N=296	**0.408* [0.232] N=181**
Mathematics	3.351	-0.129 [0.337] N=720	-0.082 [0.346] N=572	-0.145 [0.368] N=435	-0.133 [0.415] N=305	-0.005 [0.529] N=183
Science	3.521	0.073 [0.316] N=693	0.082 [0.318] N=550	0.035 [0.347] N=425	0.048 [0.372] N=300	0.184 [0.528] N=183
Foreign language	3.381	-0.454 [0.319] N=710	-0.513 [0.317] N=564	-0.560 [0.340] N=430	-0.401 [0.366] N=302	0.091 [0.455] N=183
Culture arts and sports	3.814	-0.135 [0.487] N=659	-0.076 [0.490] N=517	-0.122 [0.473] N=394	0.042 [0.558] N=273	0.231 [0.591] N=163
Vocational and technological	3.750	**-0.607** [0.271] N=521**	**-0.556* [0.294] N=411**	**-0.531* [0.316] N=309**	-0.498 [0.374] N=218	-0.117 [0.398] N=131

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.004 [0.323] N=717	0.074 [0.307] N=567	-0.047 [0.350] N=428	0.053 [0.382] N=296	0.527 [0.466] N=171
Social studies and life sciences	-0.174 [0.229] N=678	-0.112 [0.238] N=534	-0.325 [0.267] N=404	-0.227 [0.292] N=280	0.369 [0.322] N=165
Mathematics	-0.049 [0.437] N=705	0.025 [0.442] N=557	-0.048 [0.479] N=420	-0.029 [0.556] N=290	0.282 [0.780] N=168
Science	-0.138 [0.345] N=678	-0.113 [0.365] N=535	-0.188 [0.405] N=410	-0.145 [0.454] N=285	0.051 [0.692] N=168
Foreign language	-0.547 [0.377] N=696	-0.574 [0.373] N=550	-0.630 [0.398] N=416	-0.376 [0.419] N=288	0.181 [0.541] N=169
Culture arts and sports	-0.010 [0.600] N=645	0.113 [0.553] N=503	0.047 [0.522] N=380	0.295 [0.622] N=259	0.481 [0.699] N=149
Vocational and technological	**-0.905*** [0.286] N=511**	**-0.818*** [0.289] N=401**	**-0.835** [0.335] N=299**	**-0.741* [0.413] N=208**	-0.209 [0.471] N=121

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.633	-0.107 [0.089] N=1,470	-0.126 [0.091] N=1,316	-0.148 [0.095] N=1,125	-0.150 [0.102] N=936	-0.098 [0.107] N=751
Social studies and life sciences	3.656	-0.138 [0.103] N=1,398	-0.146 [0.105] N=1,254	-0.175 [0.109] N=1,071	-0.154 [0.117] N=892	-0.122 [0.118] N=713
Mathematics	3.353	0.044 [0.097] N=1,457	0.017 [0.098] N=1,302	0.002 [0.098] N=1,112	0.044 [0.106] N=925	0.069 [0.104] N=740
Science	3.543	-0.038 [0.094] N=1,395	-0.050 [0.098] N=1,246	-0.094 [0.103] N=1,065	-0.045 [0.109] N=884	0.014 [0.114] N=708
Foreign language	3.416	0.038 [0.093] N=1,435	0.028 [0.095] N=1,283	0.031 [0.099] N=1,094	0.049 [0.112] N=914	0.090 [0.111] N=734
Culture arts and sports	3.856	-0.042 [0.095] N=1,332	-0.057 [0.097] N=1,188	-0.036 [0.100] N=1,010	0.012 [0.113] N=840	-0.020 [0.122] N=673
Vocational and technological	3.753	-0.034 [0.088] N=1,002	-0.067 [0.087] N=890	-0.093 [0.091] N=764	-0.137 [0.099] N=647	-0.062 [0.109] N=528

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.137 [0.128] N=1,263	-0.147 [0.131] N=1,109	-0.153 [0.141] N=918	-0.120 [0.168] N=729	0.103 [0.197] N=544
Social studies and life sciences	-0.078 [0.156] N=1,202	-0.070 [0.156] N=1,058	-0.088 [0.161] N=875	-0.023 [0.184] N=696	0.170 [0.192] N=517
Mathematics	0.173 [0.154] N=1,254	0.143 [0.158] N=1,099	0.164 [0.163] N=909	0.273 [0.192] N=722	**0.509** [0.210] N=537**
Science	-0.084 [0.124] N=1,200	-0.091 [0.126] N=1,051	-0.144 [0.130] N=870	-0.036 [0.153] N=689	0.187 [0.143] N=513
Foreign language	-0.083 [0.174] N=1,235	-0.071 [0.176] N=1,083	-0.053 [0.183] N=894	-0.038 [0.238] N=714	0.134 [0.238] N=534
Culture arts and sports	-0.026 [0.153] N=1,147	-0.054 [0.158] N=1,003	-0.009 [0.167] N=825	0.057 [0.203] N=655	-0.139 [0.227] N=488
Vocational and technological	0.049 [0.183] N=851	0.017 [0.182] N=739	-0.029 [0.193] N=613	-0.173 [0.234] N=496	0.067 [0.327] N=377

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.633	-0.098 [0.089] N=1,470	-0.113 [0.091] N=1,316	-0.135 [0.094] N=1,125	-0.143 [0.102] N=936	-0.092 [0.107] N=751
Social studies and life sciences	3.656	-0.130 [0.102] N=1,398	-0.134 [0.104] N=1,254	-0.166 [0.108] N=1,071	-0.147 [0.115] N=892	-0.118 [0.118] N=713
Mathematics	3.353	0.047 [0.098] N=1,457	0.022 [0.098] N=1,302	0.005 [0.098] N=1,112	0.045 [0.106] N=925	0.071 [0.105] N=740
Science	3.543	-0.038 [0.094] N=1,395	-0.047 [0.097] N=1,246	-0.092 [0.103] N=1,065	-0.046 [0.109] N=884	0.011 [0.114] N=708
Foreign language	3.416	0.045 [0.091] N=1,435	0.038 [0.093] N=1,283	0.041 [0.096] N=1,094	0.054 [0.109] N=914	0.092 [0.108] N=734
Culture arts and sports	3.856	-0.039 [0.094] N=1,332	-0.053 [0.096] N=1,188	-0.031 [0.099] N=1,010	0.014 [0.111] N=840	-0.022 [0.118] N=673
Vocational and technological	3.753	-0.034 [0.087] N=1,002	-0.066 [0.086] N=890	-0.089 [0.090] N=764	-0.138 [0.098] N=647	-0.058 [0.108] N=528

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.131 [0.121] N=1,263	-0.138 [0.124] N=1,109	-0.146 [0.135] N=918	-0.126 [0.166] N=729	0.090 [0.187] N=544
Social studies and life sciences	-0.076 [0.147] N=1,202	-0.064 [0.147] N=1,058	-0.086 [0.155] N=875	-0.028 [0.180] N=696	0.156 [0.182] N=517
Mathematics	0.175 [0.155] N=1,254	0.146 [0.159] N=1,099	0.165 [0.164] N=909	0.274 [0.191] N=722	**0.504** [0.211] N=537**
Science	-0.086 [0.125] N=1,200	-0.090 [0.127] N=1,051	-0.146 [0.132] N=870	-0.041 [0.155] N=689	0.174 [0.143] N=513
Foreign language	-0.081 [0.174] N=1,235	-0.066 [0.175] N=1,083	-0.049 [0.184] N=894	-0.051 [0.241] N=714	0.113 [0.227] N=534
Culture arts and sports	-0.025 [0.154] N=1,147	-0.052 [0.157] N=1,003	-0.008 [0.167] N=825	0.055 [0.204] N=655	-0.160 [0.227] N=488
Vocational and technological	0.049 [0.183] N=851	0.018 [0.182] N=739	-0.026 [0.192] N=613	-0.175 [0.234] N=496	0.070 [0.325] N=377

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.633	-0.101 [0.089] N=1,470	-0.118 [0.090] N=1,316	-0.136 [0.094] N=1,125	-0.146 [0.101] N=936	-0.104 [0.108] N=751
Social studies and life sciences	3.656	-0.132 [0.102] N=1,398	-0.138 [0.104] N=1,254	-0.168 [0.108] N=1,071	-0.152 [0.115] N=892	-0.130 [0.120] N=713
Mathematics	3.353	0.037 [0.097] N=1,457	0.010 [0.098] N=1,302	-0.007 [0.096] N=1,112	0.031 [0.106] N=925	0.056 [0.105] N=740
Science	3.543	-0.039 [0.093] N=1,395	-0.048 [0.097] N=1,246	-0.093 [0.102] N=1,065	-0.049 [0.108] N=884	0.001 [0.116] N=708
Foreign language	3.416	0.036 [0.088] N=1,435	0.025 [0.091] N=1,283	0.031 [0.095] N=1,094	0.053 [0.106] N=914	0.080 [0.110] N=734
Culture arts and sports	3.856	-0.034 [0.091] N=1,332	-0.048 [0.094] N=1,188	-0.027 [0.097] N=1,010	0.020 [0.109] N=840	-0.017 [0.116] N=673
Vocational and technological	3.753	-0.027 [0.088] N=1,002	-0.066 [0.088] N=890	-0.085 [0.093] N=764	-0.131 [0.101] N=647	-0.055 [0.112] N=528

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.124 [0.115] N=1,263	-0.138 [0.119] N=1,109	-0.137 [0.128] N=918	-0.110 [0.157] N=729	0.086 [0.183] N=544
Social studies and life sciences	-0.065 [0.143] N=1,202	-0.059 [0.143] N=1,058	-0.070 [0.148] N=875	-0.011 [0.170] N=696	0.154 [0.184] N=517
Mathematics	0.164 [0.153] N=1,254	0.124 [0.157] N=1,099	0.153 [0.156] N=909	0.253 [0.182] N=722	**0.465** [0.205] N=537**
Science	-0.073 [0.120] N=1,200	-0.084 [0.123] N=1,051	-0.124 [0.124] N=870	-0.018 [0.144] N=689	0.175 [0.142] N=513
Foreign language	-0.065 [0.158] N=1,235	-0.063 [0.160] N=1,083	-0.035 [0.163] N=894	-0.015 [0.215] N=714	0.125 [0.221] N=534
Culture arts and sports	-0.012 [0.150] N=1,147	-0.043 [0.155] N=1,003	0.004 [0.167] N=825	0.078 [0.202] N=655	-0.126 [0.228] N=488
Vocational and technological	0.056 [0.181] N=851	0.017 [0.182] N=739	-0.027 [0.194] N=613	-0.169 [0.235] N=496	0.052 [0.327] N=377

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.633	-0.100 [0.090] N=1,470	-0.114 [0.091] N=1,316	-0.130 [0.094] N=1,125	-0.138 [0.101] N=936	-0.095 [0.109] N=751
Social studies and life sciences	3.656	-0.135 [0.103] N=1,398	-0.136 [0.104] N=1,254	-0.166 [0.107] N=1,071	-0.145 [0.115] N=892	-0.124 [0.120] N=713
Mathematics	3.353	0.044 [0.098] N=1,457	0.020 [0.099] N=1,302	0.003 [0.097] N=1,112	0.041 [0.106] N=925	0.061 [0.106] N=740
Science	3.543	-0.040 [0.095] N=1,395	-0.047 [0.098] N=1,246	-0.090 [0.103] N=1,065	-0.041 [0.108] N=884	0.009 [0.116] N=708
Foreign language	3.416	0.049 [0.088] N=1,435	0.043 [0.091] N=1,283	0.052 [0.095] N=1,094	0.076 [0.106] N=914	0.098 [0.109] N=734
Culture arts and sports	3.856	-0.043 [0.090] N=1,332	-0.054 [0.092] N=1,188	-0.029 [0.095] N=1,010	0.022 [0.107] N=840	-0.024 [0.113] N=673
Vocational and technological	3.753	-0.038 [0.085] N=1,002	-0.067 [0.084] N=890	-0.087 [0.089] N=764	-0.125 [0.095] N=647	-0.065 [0.107] N=528

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.107 [0.112] N=1,263	-0.108 [0.115] N=1,109	-0.112 [0.125] N=918	-0.066 [0.152] N=729	0.145 [0.181] N=544
Social studies and life sciences	-0.052 [0.140] N=1,202	-0.032 [0.139] N=1,058	-0.055 [0.145] N=875	0.026 [0.165] N=696	0.205 [0.175] N=517
Mathematics	0.189 [0.151] N=1,254	0.164 [0.155] N=1,099	0.183 [0.158] N=909	0.304 [0.184] N=722	**0.517** [0.213] N=537**
Science	-0.068 [0.118] N=1,200	-0.064 [0.119] N=1,051	-0.120 [0.123] N=870	0.006 [0.141] N=689	0.203 [0.142] N=513
Foreign language	-0.036 [0.157] N=1,235	-0.014 [0.156] N=1,083	0.013 [0.160] N=894	0.050 [0.210] N=714	0.189 [0.207] N=534
Culture arts and sports	-0.004 [0.145] N=1,147	-0.022 [0.149] N=1,003	0.027 [0.161] N=825	0.121 [0.196] N=655	-0.085 [0.217] N=488
Vocational and technological	0.059 [0.171] N=851	0.039 [0.171] N=739	0.001 [0.183] N=613	-0.097 [0.228] N=496	0.084 [0.315] N=377

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.622	-0.078 [0.087] N=1,393	-0.100 [0.088] N=1,249	-0.113 [0.092] N=1,069	-0.110 [0.100] N=891	-0.106 [0.111] N=714
Social studies and life sciences	3.647	-0.087 [0.107] N=1,322	-0.092 [0.109] N=1,187	-0.114 [0.113] N=1,015	-0.084 [0.120] N=847	-0.107 [0.128] N=676
Mathematics	3.345	0.075 [0.098] N=1,380	0.058 [0.099] N=1,235	0.032 [0.099] N=1,056	0.073 [0.107] N=880	0.056 [0.110] N=703
Science	3.533	0.020 [0.095] N=1,321	0.011 [0.099] N=1,180	-0.042 [0.104] N=1,009	0.008 [0.110] N=839	0.019 [0.119] N=671
Foreign language	3.406	0.082 [0.089] N=1,359	0.074 [0.092] N=1,217	0.091 [0.093] N=1,039	0.103 [0.102] N=869	0.077 [0.105] N=697
Culture arts and sports	3.853	-0.053 [0.095] N=1,269	-0.062 [0.098] N=1,133	-0.038 [0.101] N=964	0.012 [0.114] N=803	-0.022 [0.122] N=643
Vocational and technological	3.752	-0.044 [0.093] N=959	-0.084 [0.094] N=850	-0.107 [0.099] N=731	-0.142 [0.107] N=620	-0.064 [0.115] N=507

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.051 [0.116] N=1,195	-0.069 [0.120] N=1,051	-0.056 [0.127] N=871	0.003 [0.151] N=693	0.066 [0.195] N=516
Social studies and life sciences	0.004 [0.140] N=1,135	0.015 [0.140] N=1,000	0.023 [0.145] N=828	0.126 [0.159] N=660	0.113 [0.184] N=489
Mathematics	0.235 [0.159] N=1,186	0.215 [0.165] N=1,041	0.226 [0.166] N=862	**0.364* [0.188] N=686**	**0.460** [0.230] N=509**
Science	0.015 [0.130] N=1,135	0.013 [0.134] N=994	-0.045 [0.135] N=823	0.084 [0.146] N=653	0.146 [0.170] N=485
Foreign language	-0.013 [0.156] N=1,168	-0.003 [0.158] N=1,026	0.051 [0.158] N=848	0.070 [0.204] N=678	0.050 [0.215] N=506
Culture arts and sports	-0.019 [0.161] N=1,092	-0.029 [0.165] N=956	0.026 [0.176] N=787	0.078 [0.213] N=626	-0.164 [0.237] N=466
Vocational and technological	0.071 [0.189] N=813	0.039 [0.190] N=704	0.004 [0.199] N=585	-0.152 [0.241] N=474	0.069 [0.336] N=361

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.622	-0.070 [0.088] N=1,393	-0.090 [0.089] N=1,249	-0.102 [0.092] N=1,069	-0.102 [0.100] N=891	-0.095 [0.110] N=714
Social studies and life sciences	3.647	-0.087 [0.107] N=1,322	-0.093 [0.109] N=1,187	-0.115 [0.112] N=1,015	-0.091 [0.120] N=847	-0.109 [0.126] N=676
Mathematics	3.345	0.075 [0.099] N=1,380	0.055 [0.100] N=1,235	0.034 [0.099] N=1,056	0.071 [0.107] N=880	0.058 [0.110] N=703
Science	3.533	0.016 [0.096] N=1,321	0.006 [0.100] N=1,180	-0.046 [0.104] N=1,009	-0.001 [0.109] N=839	0.013 [0.117] N=671
Foreign language	3.406	0.089 [0.093] N=1,359	0.080 [0.095] N=1,217	0.100 [0.095] N=1,039	0.111 [0.104] N=869	0.091 [0.108] N=697
Culture arts and sports	3.853	-0.042 [0.097] N=1,269	-0.052 [0.100] N=1,133	-0.028 [0.102] N=964	0.015 [0.116] N=803	-0.014 [0.122] N=643
Vocational and technological	3.752	-0.033 [0.092] N=959	-0.072 [0.092] N=850	-0.094 [0.097] N=731	-0.139 [0.107] N=620	-0.061 [0.117] N=507

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.057 [0.117] N=1,195	-0.076 [0.120] N=1,051	-0.058 [0.127] N=871	0.004 [0.152] N=693	0.091 [0.194] N=516
Social studies and life sciences	0.001 [0.139] N=1,135	0.008 [0.139] N=1,000	0.018 [0.143] N=828	0.108 [0.159] N=660	0.146 [0.184] N=489
Mathematics	0.226 [0.157] N=1,186	0.206 [0.162] N=1,041	0.233 [0.164] N=862	**0.375** [0.182] N=686**	**0.486** [0.222] N=509**
Science	0.002 [0.127] N=1,135	-0.000 [0.129] N=994	-0.057 [0.132] N=823	0.075 [0.140] N=653	0.159 [0.159] N=485
Foreign language	-0.032 [0.157] N=1,168	-0.022 [0.158] N=1,026	0.034 [0.158] N=848	0.049 [0.206] N=678	0.059 [0.219] N=506
Culture arts and sports	-0.009 [0.158] N=1,092	-0.028 [0.163] N=956	0.026 [0.172] N=787	0.071 [0.212] N=626	-0.140 [0.236] N=466
Vocational and technological	0.076 [0.185] N=813	0.046 [0.185] N=704	0.010 [0.194] N=585	-0.147 [0.242] N=474	0.093 [0.336] N=361

Notes: Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.